SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 9**
- **Problem Statement Title-**

  AI-Based Crop Disease Detection and Yield Prediction System
- **Theme- AgriTech**
- **PS Category- Software**
- **Team ID- 2**
- **Team Name EventLoop**

EVENT LOOP

IDEA TITLE

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Idea / Solution

Implementation of an ***AI-powered smart farming platform*** for intelligent crop monitoring and decision-making.

- ◆ ***AI-based crop disease detection*** using crop images
- ❖ **ML-based yield prediction** using farm & crop data
- ❖ **Gemini-powered AI assistant** for agricultural guidance
- ❖ ***Real-time weather & agriculture insights***
- ❖ ***Hindi-based recommendations*** for better accessibility

Problem Resolution:

- ◆ **Early disease detection** reduces crop loss and improves timely intervention.
- ❖ **Yield prediction** helps farmers plan resources and harvest decisions.
- ❖ ***Unified platform*** eliminates the need for multiple disconnected tools.

Unique Value Propositions (UVP):

- ◆ ***AI-powered Disease → Diagnosis → Recommendation***
- ❖ **Farm-specific intelligent insights** using crop, soil & location data
- ◆ ***Multimodal AI*** combining Computer Vision, ML & Gen AI
- ❖ ***Actionable recommendations in Hindi*** for farmer accessibility

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TECHNICAL APPROACH

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AI & ML DEVELOPMENT:

Disease Detection: TensorFlow/Keras — Image-based crop disease classification.

Yield Prediction: Scikit-learn — Crop yield prediction using farm & disease data.

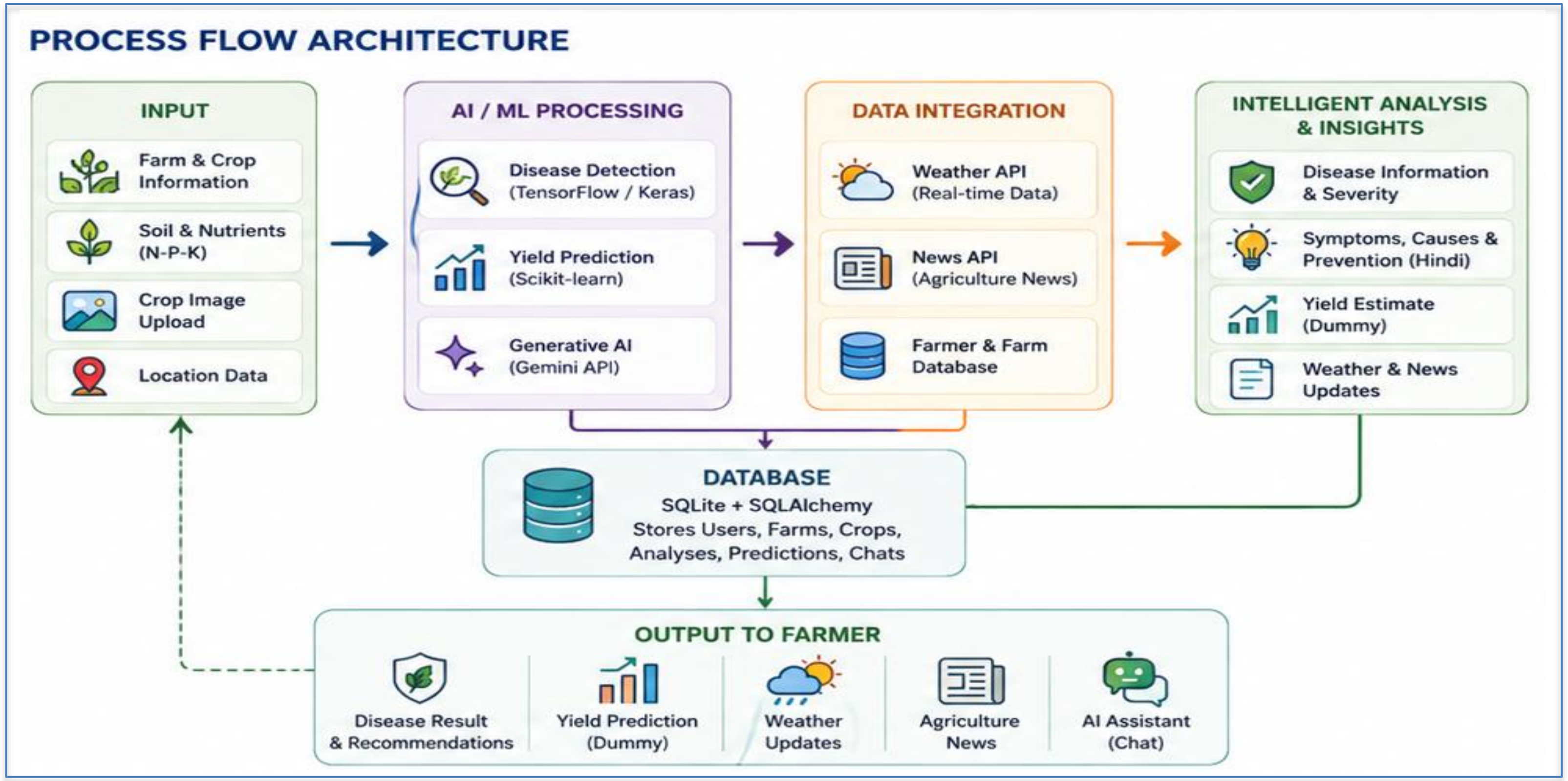
Generative AI: Gemini API — Disease insights, recommendations & AI assistance.

WEB APPLICATION DEVELOPMENT:

Backend: Flask — REST APIs, authentication & AI integration.

Frontend: HTML, CSS & JavaScript — Mobile Phone Responsive farmer interface. *****(PWA)***

Database: SQLite + SQLAlchemy — Farm, crop & prediction data management.



Product Status

~85% Product Completed — Core AI, Farm Management, Weather, News & Image Analysis modules are implemented and functional. **Yield Prediction is currently in Demo/Dummy mode**; testing and final model integration are in progress. ***** Progressive Web App***

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FEASIBILITY AND VIABILITY

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TECHNICAL FEASIBILITY

- Built using **Flask, TensorFlow, Scikit-learn & Gemini API**
- Modular architecture enables easy integration and scaling
- Lightweight web platform accessible on smartphones

OPERATIONAL FEASIBILITY

- Simple **image-based disease detection** for farmers
- Hindi AI recommendations improve accessibility
- Centralized farm, crop, weather & disease information

ECONOMIC VIABILITY

- Uses **open-source technologies** to minimize development cost
- Cloud/API-based architecture supports scalable deployment
- Can reduce crop losses through **early detection & better decisions**

SCALABILITY & FUTURE SCOPE

- Extend to **more crops, diseases & regional languages**
- Integrate IoT sensors, satellite data & advanced yield models
- Progressive Web App enables wider rural accessibility

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IMPACT AND BENEFITS

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🌾 FARMER IMPACT

- **Early disease detection** → reduces potential crop losses
- **Better yield planning** through AI-based predictions
- **Hindi guidance** makes technology more accessible

💰 ECONOMIC BENEFITS

- Reduces unnecessary crop treatment costs
- Improves resource utilization and farm productivity
- Supports data-driven farming decisions

🌱 AGRICULTURAL IMPACT

- Promotes **smart & sustainable farming**
- Enables timely weather and crop-related decisions
- Helps farmers move from reactive to **preventive crop management**

🚀 LONG-TERM IMPACT

- Scalable across **crops, regions and languages**
- Foundation for IoT, satellite & precision-agriculture integration

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RESEARCH AND REFERENCES

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1. **Mohanty, S. P., Hughes, D. P. & Salathé, M. (2016)**
   Using Deep Learning for Image-Based Plant Disease Detection
   → Foundation for AI-based crop disease detection using images.

2. **van Klompenburg, T., Kassahun, A. & Catal, C. (2020)**
   Crop Yield Prediction Using Machine Learning: A Systematic Literature Review
   → Supports ML-based crop yield prediction using agricultural factors.

3. **Upadhyay, A. et al. (2025)**
   Deep Learning and Computer Vision in Plant Disease Detection
   → Recent research on computer vision and deep learning in precision agriculture.

4. **Jabed, M. A. & Murad, M. A. A. (2024)**
   Crop Yield Prediction in Agriculture: A Comprehensive Review of ML & DL Approaches
   → Supports data-driven yield prediction and sustainable agriculture.